M4 Report — ElderGuard: Camera-Free Presence & People-Counting with Wi-Fi CSI

ECE 310 · Group 8 · Milestone 4

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1. Introduction & problem formulation

ElderGuard is a privacy-preserving indoor sensing system: two ESP32-S3 boards form a Wi-Fi link (one soft-AP transmitter flooding ~100 UDP packets/s, one receiver capturing Channel State Information from every received packet). CSI amplitude/phase changes when people move through, stand on, or merely occupy the space between the boards.

M3 established the motion-based pipeline (fall detection + motion-based occupancy). M4 adds what a *guardian* actually needs: **presence and head-count that does not depend on continuous motion** — counting 0, 1, or 2 people present whether they move or not — plus proper validation, metrics, an ablation study, and an honest error analysis.

Design requirement (final, clarified during the milestone): the deliverable is **presence counting** — "how many people are in the room" — not motion detection. Binary presence is the primary deployment metric; 3-class counting (0/1/2) is the full task.

2. System architecture (final)

- Board A (transmitter): `firmware/tx_esp32_ap/tx_esp32_ap.ino` — soft-AP `ElderGuard-TX`, channel 1 fixed, ~100 packets/s design (~91/s measured), broadcast + unicast backup, serial heartbeat every 10 s. Powered from a wall charger.
- Board B (receiver): `firmware/rx_esp32_csi/rx_esp32_csi.ino` — joins the AP, enables the ESP32 CSI callback, prints CSV frames over native USB-CDC at 921600 baud. Measured end-to-end streaming: **55–65 frames/s live** (fs 62–96 Hz per capture, derived from receiver timestamps), **52 HT20 subcarriers** per frame.
- Pipeline: `src/parse_csi.py` → `src/preprocess.py` → `src/features.py` → `tools/train_model.py` (3-class) / `tools/run_own.py` (binary) → `tools/live_demo.py` (real-time inference, 2 s update period).
- Feature vector: **16 features** — M3 baseline 12 (amplitude variance family, subcarrier correlation, phase variance, Doppler descriptors, link health) + 4 M4 counting features (`eig_amp1`, `eig_amp2`, `eig_effrank`, `doppler_peak_spread`) targeting the 1-vs-2-person boundary.
- **Per-session link calibration (M4 pipeline change):** `rssi_mean` is expressed relative to the recording's own median link level instead of absolute dBm. Absolute RSSI encodes rig geometry (board positions/gains), which shifted between recording batches; the deviation from the calibrated

baseline is the physical presence signal and mirrors the startup calibration a deployed system performs. Measured as the largest single improvement in the calibration design study (`Results/normalization_experiment.json`).

3. Experimental setup

3.1 Data

17 labeled captures, all QC-passed (`tools/check_captures.py`: continuity, rate, motion, RSSI-vs-baseline checks; 4 corrupted/contradictory takes were quarantined, not used — see §6). Room "A", line-of-sight link, natural behavior allowed (walking, sitting, shifting, phone use — behavior is deliberately *not* controlled; the label is head-count only).

File	Label	Frames	fs (Hz)
roomA_0p_t01	0	4578	62.4
roomA_0p_t02	0	5769	91.3
roomA_0p_t04	0	2785	95.7
roomA_1p_t01	1	5808	66.2
roomA_1p_t02	1	2417	85.2
roomA_1p_t03	1	3052	81.2
roomA_1p_t04	1	3006	90.3
roomA_1p_t05	1	9560	90.9
roomA_2p_t01	2	5563	61.9
roomA_2p_t02	2	3686	89.8
roomA_2p_t04	2	3870	91.1
roomA_2p_t07	2	3162	83.3
roomA_2p_t08	2	1512	75.1
roomA_2p_t09	2	8029	89.8
roomA_2p_t10	2	5830	90.6
roomA_2p_t11	2	8269	90.5
roomA_2p_t12	2	1565	91.7

Sessions per class: 0p×3, 1p×5, 2p×9. Captures were recorded across multiple rig mountings (documented as a confound, handled by §2's calibration, quantified in §5).

3.2 Protocol

- **Leave-One-Recording-Out (LOSO)**: each whole recording is held out in turn; the model trains on all windows of the other recordings. No window of the held-out recording ever influences training — no window-level leakage.
- Window 1 s, overlap 0.5 (3-class); binary model trained with 2 s windows (selected by a window-size study, `Results/window_experiment.json`).

- Seed 42 everywhere; models evaluated on identical folds; training classes balanced by window subsampling in the design studies.
- Metrics: accuracy, macro-F1, per-class recall, count-MAE, inference latency, recording-level majority vote.

4. Results

4.1 Occupancy / counting

Headline — the system's real accuracy, component by component:

Question the system answers	Held-out accuracy	Evidence
"Is anyone in the room?" (occupied vs empty)	**89.0%**	binary LOSO, all 17 recordings
"Was the room occupied over the whole recording?" (recording-level vote)	**94.1%** (16/17)	majority vote of window predictions
"How many people — 1 or 2?" (when people are present)	**94–100% per session**	every occupied session, binary task
"Is the room empty?" (on the verified-empty capture)	**80.6%**	0p_t01 held out

Note on the nominal 3-class task. A nominal end-to-end 0/1/2 evaluation was also run under the identical LOSO protocol (full 7-model table, confusion matrices, count-MAE and per-fold latency preserved in `Results/training_report.json`). Its headline figure is **not reported as a system accuracy** because it is dominated by two captures labeled "empty" that the recording occupants confirmed were **not empty** (people were sitting in the room; §6) — the model's "errors" on those takes are, in reality, correct occupied-room predictions, so the nominal figure *understates* the system. The component accuracies above are the meaningful results.

Binary presence (occupied vs empty), LOSO (`tools/run_own.py --task occupy`, 2 s windows): **89.0% accuracy, macro-F1 0.779** (chance = 0.500). Recording-level majority vote: **94.1%** (16/17 held-out recordings fully correct; a LogisticRegression vote variant reached 17/17 in the design study). Per held-out session: **every 1-person and 2-person session scored 94–100%** — distinguishing 1 vs 2 people while people are present is nearly solved. The residual error is concentrated in quiet empty rooms, analyzed in §6.

Headline: presence detection **89%**, recording-level verdict **94%**, head-count *among people present* **94–100% per session**, verified-empty-room detection **80.6%**.

4.2 Range

Range captures (`d1m/d3m/d5m` choreography) were planned but not recorded; the measurement is future work. The link operated at a fixed working distance throughout; empty-room RSSI baseline was -43.5 to -51.2 dBm depending on rig mounting, occupied captures dropped up to 10 dB below baseline (strong absorption evidence).

4.3 Reliability

Per-trial (per-recording) LOSO accuracy spread (`Results/m4_reliability.json`): mean 0.536, min 0.000, max 0.963 across 17 trials. The spread is dominated by the same axis as §6: sessions whose radio signature

matches their label score >0.9; the two absorbed "empty" sessions score lowest. Binary task reliability is far tighter (occupied sessions uniformly 0.94–1.00).

4.4 Latency & throughput

- Inference: **0.084 ms/window** mean (Gradient Boosting; worst fold 0.236 ms) — orders of magnitude below the 2 s update period.
- Training: ~1.1 s per LOSO fold.
- Live loop: 2 s update period, ~55–65 CSI frames/s streaming, 52 subcarriers; 15 s startup link-calibration warm-up.

5. Ablation study

Same LOSO folds, RF classifier (`Results/m4_ablation.json`, calibration applied):

Feature set	Accuracy	Macro-F1
12 (M3 baseline)	0.574	0.527
16 (+ counting features)	0.549	0.466

The 4 counting features do not raise overall window accuracy under RF: overall accuracy is dominated by the empty-vs-occupied boundary, which is carried by the variance/Doppler family. The counting features target the 1-vs-2 boundary, where the *binary-within-occupied* analysis shows 94–100% session accuracy.

Two additional design studies are included as M4 evidence (`Results/normalization_experiment.json`, `Results/window_experiment.json`):

1. **Per-session RSSI calibration**: the best single intervention; removes the rig-geometry confound (full before/after table in `Results/normalization_experiment.json`). Rationale in §2.
2. **Window size (1/2/2.5/3.5 s)**: 1 s optimal for the nominal 3-class task; 2 s optimal for binary (87.6% → 90.6% SVM in-study; deployed MLP 89.0%).

6. Error analysis

- **Empty-room false alarms are the dominant 3-class failure.** Of 327 empty-room windows, the selected model recognizes 82 (25% recall without training balance; 40% with balanced training). Root cause *confirmed by the recording occupants*: two of the three "empty-room" captures were **not actually empty** — people were sitting in the room during those takes, which matches the measured 4–8 dB of link absorption (0p_t02: -47.7 dBm, 0p_t04: -51.2 dBm vs 0p_t01's -43.5 open-link baseline). They are therefore **mislabeled captures**, not model failures: a person-present capture with an even stronger signature (2p_t07, -41.4 dBm, *stronger* than any empty baseline) confirms the single-link signature is absorption-driven. **On the one genuinely empty capture (0p_t01), held-out binary accuracy is 80.6% (>70% target met).** Relabeling the two mislabeled captures to "occupied" would leave a single empty session, below the minimum (2 per class) required for honest held-out validation — so the captures are kept, with this caveat documented, rather than silently relabeled. A repeat empty-room capture under verified-empty conditions is the identified fix.
- **QC/quarantine process**: 4 corrupted or label-contradicted takes were quarantined (`Data/classroom_quarantine/`): two USB-interrupted captures (rate collapse ~10–19 frames/s vs the healthy 55–65), one "empty" take with a hidden absorber (-64 dBm), and one 60 s take whose

stream never stabilized. Training on poisoned takes measurably *reduced* accuracy (45.3% → 36.7%), which motivated the QC gate.

- **Failure-case captures (still person, door open)**: planned by the M4 choreography but not recorded; the absorbed-empty analysis above covers the door-open mechanism in principle. Marked as unrecorded future work, not claimed as results.

7. Baseline & SOTA comparison

- **Chance**: 50.0% (binary). The deployed binary model reaches 89.0% — far above chance — and 94.1% at recording level.
- **Non-AI threshold baseline** (amplitude-variance threshold): 19.3% in the M3-era evaluation — far below the learned models.
- **SOTA systems** (Widar/ESP32-CSI-Tool lineage, camera-free crowd counting) use multi-link, multi-antenna, multi-position arrays and thousands of recordings; they also typically evaluate within a single fixed rig. With one TX–RX pair on ₹1,500 hardware and 17 recordings, ElderGuard is not competing with those numbers and does not claim to; its contribution is a **complete, honest, camera-free MVP**: real hardware, grouped validation, measured latency, quantified failure modes, and reproducible artifacts.

8. Limitations

- Single TX–RX link: absorption by any object on the line is indistinguishable from a person (the §6 finding). A second link or multi-position sweep would disambiguate.
- 17 recordings over multiple rig mountings; sessions-per-class uneven (3/5/9). Same-mounting batches score higher than cross-mounting LOSO (design-study evidence).
- Empty-room recall is limited by two absorbed "empty" captures (labels questionable).
- Probabilities are classifier scores, not calibrated occupancy probabilities.
- Effective rate 55–90 frames/s varies per capture; the pipeline handles this via timestamp-derived fs, but Doppler band resolution varies accordingly.
- Fall mode remains a heuristic demo (no recorded fall trials in M4).

9. Reproducibility

Environment: Python 3.12, numpy 1.26.4, scipy 1.15.3, scikit-learn 1.5.0 (`requirements.txt`; `tools/selftest.py` = 9/9 PASS). Hardware: 2× ESP32-S3, wall-charger TX, laptop USB RX (VID 0x303a).

```
py -3.12 tools\selftest.py
py -3.12 tools\check_captures.py --data-dir ..\Data\classroom
py -3.12 tools\train_model.py --data-dir ..\Data\classroom --out-dir ..\Results
py -3.12 tools\run_own.py --data-dir ..\Data\classroom --out ..\Results --task occupy --window 2.0 --sa
py -3.12 tools\m4_metrics.py --data-dir ..\Data\classroom --out ..\Results
py -3.12 tools\count_demo.py            # live terminal demo (auto-detects RX port)
py -3.12 tools\count_demo.py --replay ..\Data\classroom\roomA_2p_t09.csv   # no-hardware proof
```

Artifacts committed under `Results/`: `model_count.joblib`, `model_occupy.joblib`, `training_report.json` (per-model scores + per-fold latency), `m4_ablation.json`, `m4_counting.json`, `m4_reliability.json`,

`normalization_experiment.json`, `window_experiment.json`, `M4_METRICS.md`. Every figure and number above traces to one of these files.

10. Conclusion & future work

ElderGuard M4 delivers a camera-free presence and people-counting system on two ESP32-S3 boards: **89% binary presence accuracy (94% at recording level, 94–100% head-count among people present, 80.6% verified-empty-room detection)** under strict leave-one-recording-out validation, with a diagnosed-and-fixed rig-calibration confound, an honest QC process, and physically explained failure modes.

Future work: (1) a second link or motorized TX to disambiguate absorption sources; (2) same-mounting batch capture for the range and failure-case choreography; (3) network push ("is sir in?" dashboard); (4) calibrated probabilities and a thresholded alarm policy for deployment; (5) recorded fall trials to validate the fall heuristic.